

Chapter 7 Continuity

7.1 Limits and continuity

A function f is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$.

Equivalently: for every $\epsilon > 0$ there is a $\delta > 0$ such that

$|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$.

The order of the quantifiers matters: δ is chosen after ϵ and after a .

7.1 Limits and continuity

Example. $f(x) = x^2$ is continuous at every a .

Given $\epsilon > 0$, choose $\delta = \min(1, \epsilon / (2|a| + 1))$.

Then $|x^2 - a^2| = |x - a| |x + a| < \delta (2|a| + 1) \leq \epsilon$.

Notice that δ depends on a as well as on ϵ .

7.2 Three hard theorems

Theorem 1. If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some x in $[a, b]$ with $f(x) = 0$.

Theorem 2. If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$.

Theorem 3. If f is continuous on $[a, b]$, then there is some y in $[a, b]$ such that $f(y) \geq f(x)$ for all x in $[a, b]$.